

Preliminary Design, Analysis, and Characterization of the Pointing, Acquisition, and Tracking Subsystem for the PULSE-A CubeSat

by
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ABSTRACT

Recent reductions in the size, weight, and power (SWaP) requirements of sensing and observation technologies have significantly increased the rate of space-based data collection, driving demand for higher space-to-ground bandwidth than can be achieved using traditional radio-frequency (RF) communication systems. Optical communication systems address this need, offering order-of-magnitude increases in bandwidth while providing enhanced security through narrow optical beam divergence and stringent pointing requirements. At the same time, the high information density of modern sensing and the rise in commercial launch opportunities have driven demand for compact and affordable optical communication architectures. The Polarization-modULated Laser Satellite Experiment (PULSE-A) is a 10x10x30cm CubeSat mission being developed at the University of Chicago under Professor Tian Zhong. The mission aims to demonstrate a novel optical modulation scheme, Circular Polarization Shift Keying (CPolSK), which encodes binary information in left-and right-handed circular polarization states of light. PULSE-A will provide the first on-orbit demonstration of CPolSK, testing its noise tolerance and establishing a foundation for future space-to-ground quantum optical links. This thesis will focus on the research, design, and testing of the optical and electronic systems enabling this demonstration. While an advantage for ensuring security, the narrow divergence and strict laser pointing requirements are a major challenge for all space-to-ground optical communications systems. The solution is a fine-pointing mechanism that uses a fast steering mirror (FSM) to point the transmission laser towards the optical ground station with less than 0.35 mrad precision. The second aspect of the thesis project is the research, development, characterization, and test of this pointing scheme. Extensive testing is required to understand the response of onboard photodetectors as well as the FSM, which form a closed-loop control system. These systems will be integrated into the PULSE-A payload ahead of its planned launch in early 2027, establishing the groundwork for future CubeSat-scale optical and quantum communication missions.

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1 Introduction

Space-to-ground optical communication (lasercom) has emerged as a pivotal technology in the small satellite sector. By transmitting data using light rather than radio frequency (RF) signals, lasercom achieves significantly higher bandwidth while reducing size, weight, power consumption, and cost compared to conventional RF systems [1]. These capabilities have transformative implications for low-cost space platforms such as CubeSats, potentially enabling major scientific discoveries that would otherwise require far more expensive spacecraft. Lasercom also lays the groundwork for future global quantum networks, something that terrestrial fiber networks alone cannot support [2].

The Polarization ModULated Satellite Experiment (PULSE-A), developed at the University of Chicago, is the first lasercom experiment in the world to be designed and built entirely by undergraduate students. PULSE-A aims to demonstrate 10 Mbps optical downlink using Circular Polarization Shift-Keying (CPolSK) modulation. This thesis provides an overview of lasercom on CubeSat platforms before discussing the preliminary design, development, and analysis of the Pointing, Acquisition, and Tracking (PAT) hardware subsystem for PULSE-A.

2 Motivation

Space-based science is critical to modern experimental physics, enabling observations and measurements that are fundamentally impossible to conduct from the ground. Over the past five decades, satellites and orbital observatories have revealed entirely new physical phenomena and reshaped our understanding of the universe. The Cosmic Microwave Background (CMB), mapped by the COBE and WMAP missions, was instrumental in establishing the standard cosmological model—one of the most significant scientific achievements of the 20th century.

As scientific instrumentation has grown smaller, more sensitive, and more energy-efficient, the satellite paradigm has evolved in parallel. Large flagship missions like Hubble and JWST remain worthwhile, but compact platforms play an increasingly prominent role. Small satellites (typically defined as below 500 kg) and nanosatellites such as CubeSats (1–10 kg) [3] are now routinely used to test new scientific principles, validate emerging technologies, and collect high-fidelity data on orbit.

2.1 CubeSats

CubeSats are a class of nanosatellite built around a standardized modular form factor, where each unit ("U") measures roughly 10x10x10 cm, designed to make space missions more affordable and accessible. Unlike large "flagship" spacecraft—which are typically engineered from the ground up for a specific mission—CubeSats can draw from a shared supply chain, driving costs down and improving reliability through flight-proven components.

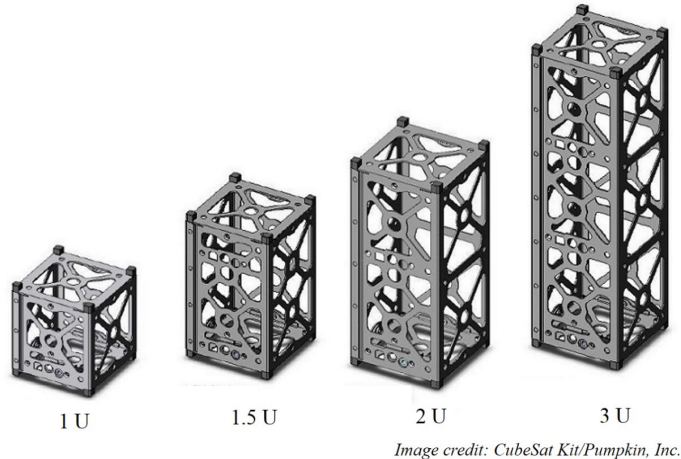


Figure 1: Common CubeSat frame sizes, from 1U to 3U.

Originally developed by California Polytechnic State University and Stanford University [4], CubeSats are now widely used for research [2] and educational projects by government agencies (such as NASA, ESA, DLR, and JAXA) and universities. This paradigm shift has transformed low-Earth orbit into an accessible experimental platform where small spacecraft can meaningfully contribute to cutting-edge research. Research at the University of Chicago alone demonstrates the impact on the accessibility of space: the Dark Matter Nano-satellite Equipped with Skipper Sensors (DarkNESS) mission aims to set competitive upper limits on Strongly-Interacting Dark Matter in a CubeSat [5], and the Climate Systems Engineering Initiative aims to build a Stratospheric Aerosol Injection monitoring CubeSat constellation. Both missions would have been practically infeasible just two decades ago without the introduction of the CubeSat.

2.2 Optical Communications

The recent reductions in the size, weight, and power (SWaP) of sensing and observation technology to fit within CubeSats have drastically increased space-based data collection rates, motivating the need for greater space-to-ground bandwidth than is achievable with traditional radio-frequency (RF) systems. In the case of DarkNESS, for example, the scientific gain is limited by how much high-quality image data can be downlinked during a single RF pass over Illinois. Optical communication systems address this demand by enabling order-of-magnitude increases in bandwidth while also improving link security through narrow beam divergence and stringent pointing requirements [6].

The intrinsic bandwidth advantage of optical communications (lasercom) over RF systems arises from the much higher carrier frequency and narrow divergence. Optical communications in the telecom C-band operate near 200 THz, compared to typical UHF systems operating near 400 MHz, enabling orders-of-magnitude greater achievable data throughput. In addition, The optical spectrum faces substantially fewer regulatory bandwidth constraints than RF

communications and provides multi-THz of usable bandwidth for communications [1].

The smallsat as a widely available tool has driven demand for compact and affordable optical communication architectures. With advancements in the field, comparable data-rate optical communication systems are often several times cheaper than their RF counterparts.

3 PULSE-A Mission Overview

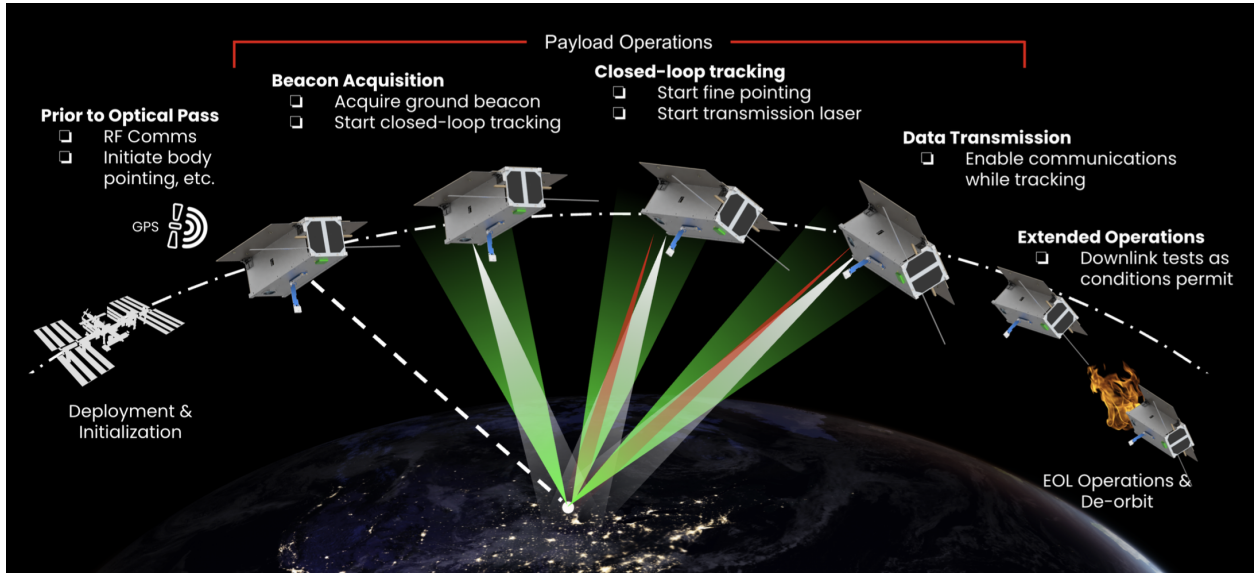


Figure 2: Simplified PULSE-A mission concept of operations, detailing the responsibilities of the space and ground segments. The green laser, pointing from ground to space, is the Optical Ground Station (OGS) beacon laser, used for tracking. The white laser, pointing from space to ground, is the Payload’s beacon laser, also used for tracking. The red laser is the spacecraft’s transmission laser, which encodes data in circularly polarized light [7].

The Polarization-modULated Laser Satellite Experiment (PULSE-A) is a 3U CubeSat mission being developed at the University of Chicago. The mission aims to demonstrate a novel optical modulation scheme, Circular Polarization Shift Keying (CPolSK), which encodes binary information in the left- and right-handed circular polarization states of light.

Most compact lasercom terminals rely on amplitude modulation, such as On–Off Keying (OOK), for its simplicity and robustness [8]. Existing CubeSat lasercom demonstrations—including the German Aerospace Center (DLR)’s OSIRIS program and the Massachusetts Institute of Technology’s CLICK CubeSats—have successfully used OOK architectures to achieve reliable downlinks within stringent SWaP constraints [9] [10]. Unlike OOK, CPolSK uses a fixed-amplitude modulation scheme, providing improved robustness in addition to the inherent advantages of polarization maintenance through the atmosphere (see Figure 3 for a visual comparison between OOK and CPolSK modulation schemes) [11]. Information is

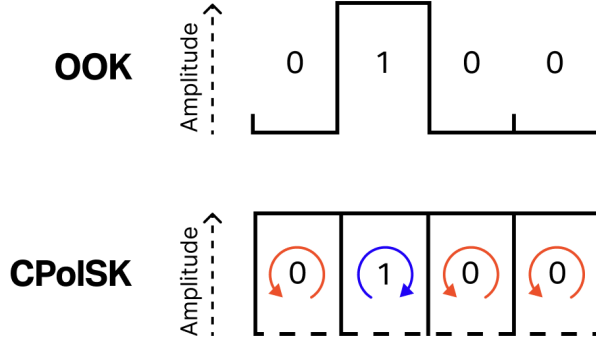


Figure 3: OOK vs. CPolSK modulation architectures. Note the constant amplitude of the CPolSK signal. *Image reproduced with permission from [6].*

encoded by switching between orthogonal polarization states, allowing the laser amplitude to remain constant for each transmission bit [12] and resulting in an approximately 3 dB lower required SNR than standard OOK, even after accounting for the conversion of polarization to intensity at the detector [6]. While orthogonal linear polarization offers similar propagation benefits and is simpler to implement, CPolSK was chosen because circular polarization states are rotationally invariant under rotation about the transmission axis, eliminating the need to correct for field rotation at the receiver and simplifying ground tracking and demodulation [6]. This rotational invariance is particularly advantageous for dynamic satellite-to-ground links, where the transmitter and receiver are in constant relative motion.

Moreover, CPolSK serves as a direct technological foundation for polarization-based quantum communication, particularly Quantum Key Distribution (QKD): the circular polarization bases generated by the CPolSK scheme can naturally serve as one of the two mutually unbiased polarization bases required by the BB84 protocol [6]. PULSE-A will serve as the first on-orbit demonstration of CPolSK [6], evaluating its noise tolerance and laying the groundwork for future space-to-ground quantum optical links.

PULSE-A’s optical terminal aims to demonstrate 10 Mbps CPolSK-encoded space-to-ground communication from Low Earth Orbit, exceeding the bandwidth performance of typical 9600 baud CubeSat radios by nearly three orders of magnitude at only five times the cost.

4 Pointing, Acquisition, and Tracking

Unlike omnidirectional RF systems, lasercom terminals must orient their optical line-of-sight with exceptional precision due to the narrow divergence of the transmission beam, a downside of the feature that gives lasercom its bandwidth advantage [2]. PULSE-A will be deployed into a 450 km circular orbit, transmitting to an Optical Ground Station (OGS) located in Chicago. Optical links will be attempted during passes in which the satellite is between 30°

and 70° above the horizon for a duration of at least two minutes. Over such a pass, the range to the OGS varies from approximately 500 km at closest approach to nearly 900 km near the edges of the link window. Given the divergence of the transmit beam, this corresponds to a beam footprint of roughly 1 km in diameter at the ground—within which the OGS must lie to close the link.

This constraint, combined with the signal-to-noise ratio (SNR) requirement of the OGS receiving hardware, results in a transmit (Tx) beam pointing requirement of 0.35 mrad (0.02°) across all axes. The available link margin—the excess SNR above the minimum threshold at the ground station—permits up to 0.35 mrad of off-boresight mispointing before the link is lost, as illustrated in Figure 4. Beyond this threshold, the received SNR falls below the minimum required, the link cannot be closed, and no data is transmitted.

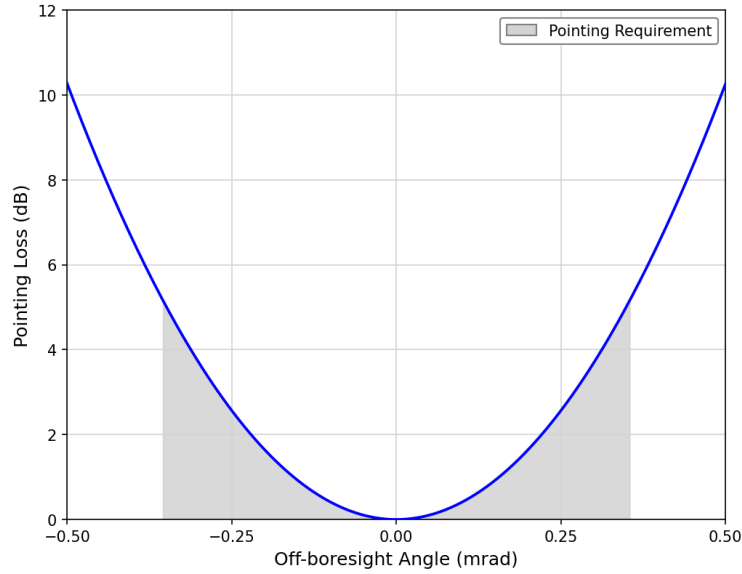


Figure 4: Quadratic dependence of pointing loss on off-boresight angle. The shaded region indicates the range of acceptable off-boresight mispointing corresponding to an SNR of 5.2 dB. Under this SNR constraint, a transmitter beam pointing requirement of 0.35 mrad is obtained.

The subsystem responsible for orienting the Tx beam and maintaining alignment throughout a pass is known as the Pointing, Acquisition, and Tracking (PAT) subsystem, and is the primary focus of this work. The following sections outline the key design considerations driving PULSE-A’s PAT architecture, before turning to methods for characterization and analysis of the system.

4.1 Coarse Pointing

Optical communications terminals, like PULSE-A, typically employ one of three methods to achieve the required pointing accuracy.

1. Ultra-high accuracy Attitude Determination and Control Systems (ADCS).

This is a new development in the CubeSat sector with recent technology improvements. Several vendors (most notably Blue Canyon [13]) can deliver ADCS units with arcsecond-order (0.01 mrad) spacecraft pointing accuracy, sufficient to directly satisfy many CubeSat lasercom pointing requirements. The downside of this approach is their significant size, weight, power, and cost.



Figure 5: Tesat-Spacecom SCOT80 Terminal with active gimbal mount to compensate for spacecraft coarse pointing error [14].

2. Gimbaledd telescopes mounted external to the spacecraft [1]. This yields less reliance on the spacecraft’s intrinsic pointing capabilities but significantly increases the complexity and size of the terminal. This is commonly used on commercial lasercom products designed to operate within a range of satellite pointing capabilities. Gimbaledd systems provide reliable high-accuracy pointing for LEO constellation crosslink architectures, such as SpaceX’s Starlink or other proposed constellations [15][16][17]. For university-class CubeSats, which often sit in the 1 to 6U range, gimbaledd terminals are simply infeasible due to the size requirements.

3. Medium-accuracy ADCS systems. Standard off-the-shelf ADCS units can now deliver 3-sigma pointing accuracies in the range of 0.5° to 1° as a standard requirement. This provides CubeSats with high-accuracy positional knowledge that was not available when CLICK-A was developed. Systems such as CubeSpace’s Gen2 ADCS can provide roughly 0.6° 3-sigma pointing accuracy for around \$50k, a level of performance that would have been inaccessible only a decade ago. These systems are now common on Earth observation, remote sensing, and most spacecraft larger than 2U, making it reasonable to design a university-class lasercom terminal around this level of body-pointing capability. However, unlike the first approach, the spacecraft alone still cannot meet the 0.35 mrad Tx beam pointing requirement.

4.2 Staged Control

PULSE-A employs the third pointing strategy using a COTS ADCS from CubeSpace Satellite Systems. The system provides roughly 11 mrad 3-sigma body-pointing control accuracy on each axis of the spacecraft-to-OGS vector, which is sufficient to place the optical line-of-sight

near the target direction but insufficient to maintain the signal-to-noise ratio required for reliable optical communication. To overcome this limitation, PULSE-A implements an additional fine-pointing stage to actively maintain precise beam alignment throughout the pass. Two-stage control is common in lasercom terminals using either gimbaled telescopes or medium-precision ADCS units [18][2]. The coarse-pointing stage may operate in either open- or closed-loop control, while the fine-pointing stage is typically closed loop to minimize pointing error [riesing2015].

PULSE-A uses open-loop coarse pointing through the spacecraft ADCS. CubeSpace provides the ground-station tracking algorithms using inputs from the onboard star tracker, inertial measurement unit, and magnetometer. However, the ADCS slew-rate limitations (the rate at which the spacecraft can change attitude) make passes directly overhead difficult due to the required angular tracking rate. This constraint motivates the 70° maximum elevation requirement discussed earlier, while the 30° minimum elevation is driven by link budget limitations.

4.3 Beacon Architecture

Optical communication systems typically rely on a ground-to-space beacon (referred to as Rx beam) for acquisition and alignment. [2]. The Rx beam is collected by the onboard aperture, allowing for the measurement of the coarse pointing error using an internal sensor. For low noise behavior as well as simplicity in algorithm development, many terminals, like PULSE-A, use a Quadrant Photodiode (QPD) [19][18] onboard to measure Rx angular error rather than a high-resolution Focal Plane Array (FPA). When the spacecraft is perfectly aligned with the uplink beacon, the spot is centered on the QPD. Any misalignment produces a displacement from the center, which can be measured by comparing the voltages across the four quadrants using the following equations:

$$x_c = \frac{(Q_1 + Q_4) - (Q_2 + Q_3)}{Q_1 + Q_2 + Q_3 + Q_4} \quad (1)$$

$$y_c = \frac{(Q_1 + Q_2) - (Q_3 + Q_4)}{Q_1 + Q_2 + Q_3 + Q_4} \quad (2)$$

PULSE-A is also unique in that it has a space-to-ground beacon that is enabled before the Tx beam. This allows the OGS to also enter a fine pointing stage to improve tracking capabilities. This path is distinct to the optical path in which the Tx and Rx beams pass.

4.4 Fine Pointing Fundamentals

This angular error can then be fed into a Fast Steering Mirror (FSM), which may use mechanical, piezoelectric, or Micro-Electro-Mechanical Systems (MEMS) actuation [21].

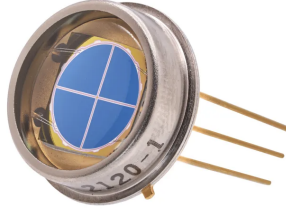


Figure 6: Quadrant Photodiode similar to the one used on the PULSE-A optical terminal. Quadrants are labeled counterclockwise from the top right [20].

Advances in MEMS technology have made MEMS FSMs particularly well suited for compact lasercom architectures, with multiple successful flight demonstrations [22].

This error is handled by a closed-loop control system that commands the tip-tilt fast steering mirror (FSM) within the optical terminal. The FSM responds at high speed, making small, continuous angular adjustments that steer the outgoing communication beam back toward the direction of the Rx beacon laser.

Through the combination of coarse and fine pointing corrections, PULSE-A is able to stabilize its Tx beam to 0.35 mrad accuracy, enabling the mission to achieve 10 Mbps optical downlink. This architecture is common in CubeSat form-factor terminals, and is visualized in Figure 7.

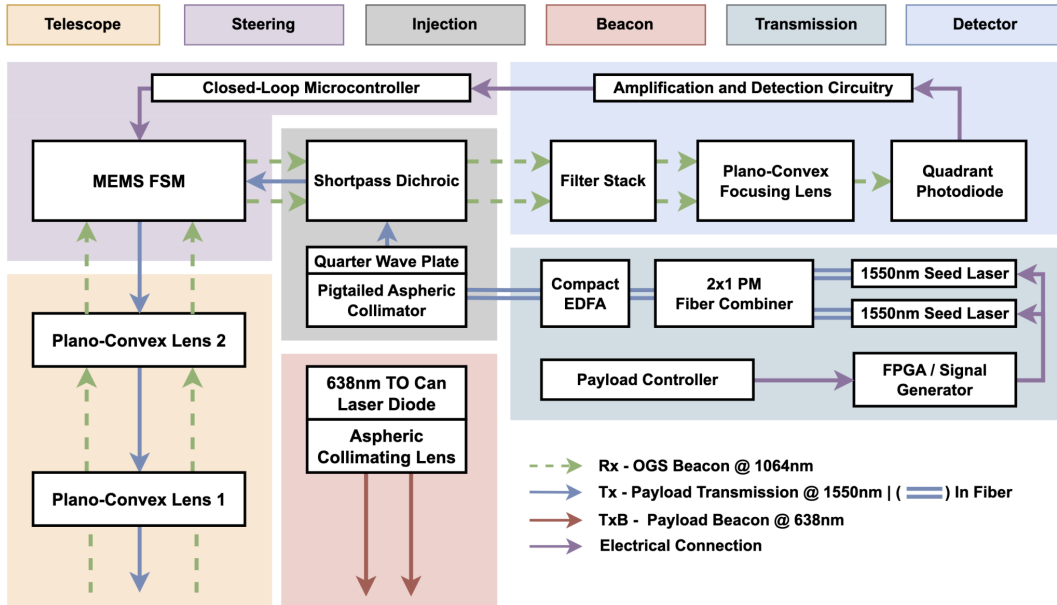


Figure 7: Optical and control architecture of the PULSE-A optical terminal. Received beacon light is focused onto a Quadrant Photodiode (QPD), whose centroid error signal drives a Fast Steering Mirror [6].

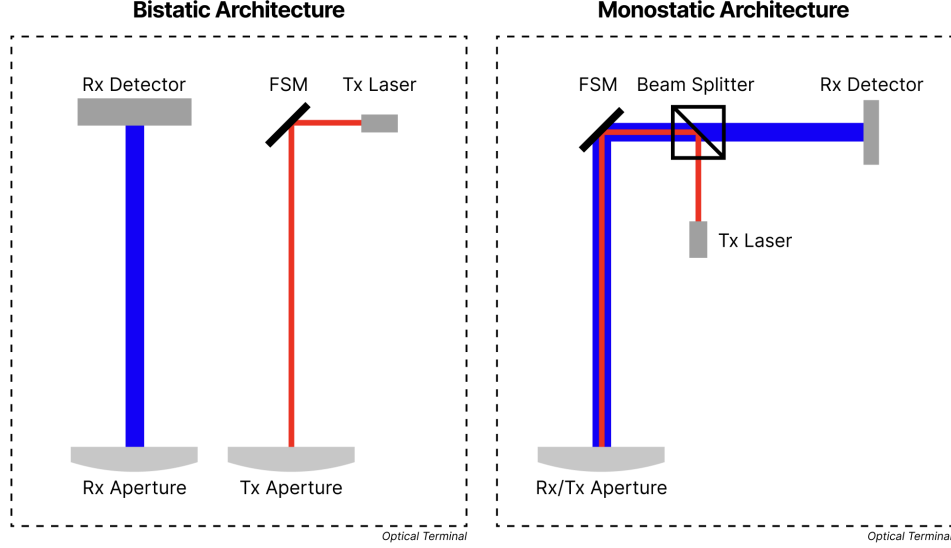


Figure 8: Bistatic vs. monostatic lasercom terminal architectures. The bistatic design (left) separates transmit and receive apertures; the monostatic design (right) combines them via a beam splitter, enabling a more compact, co-aligned optical path.

4.5 Monostatic vs Bistatic Architectures

In addition to the considerations discussed above, all lasercom systems must adopt one of two primary optical path alignment configurations: monostatic or bistatic architectures (Figure 8).

The monostatic approach enables direct closed-loop feedback control because the Tx and Rx beams are co-boresighted. This provides a compact and highly capable optical design, but requires the FSM to compensate for angular deviations amplified by the telescope’s angular magnification. As shown through basic ray analysis in Figure 9, any pointing error observed at the receive aperture appears magnified at the transmit path.

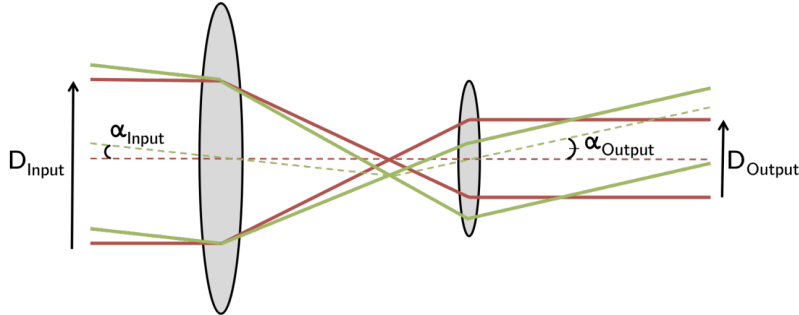


Figure 9: Angular magnification through a two-lens telescope system [riesing2015]. A beam entering at diameter D_{Input} and angle α_{Input} is compressed to D_{Output} at the output aperture, amplifying the steering angle to α_{Output} [riesing2015].

In contrast, bistatic configurations separate the Tx and Rx optical paths, preventing direct closed-loop feedback during fine pointing. While this complicates tracking performance, bistatic systems are generally simpler to develop and test because the transmit and receive apertures operate independently. Additionally, the FSM in a bistatic system does not experience the large angular amplification introduced by the receive telescope, significantly reducing the required FSM control authority [23][riesing2015].

PULSE-A selected a monostatic configuration due to advancements in both ADCS and MEMS technology, which reduce the expected spacecraft pointing error to within the control authority of the FSM despite the telescope’s 7.5x angular magnification factor.

4.6 PULSE-A PAT Architecture Overview

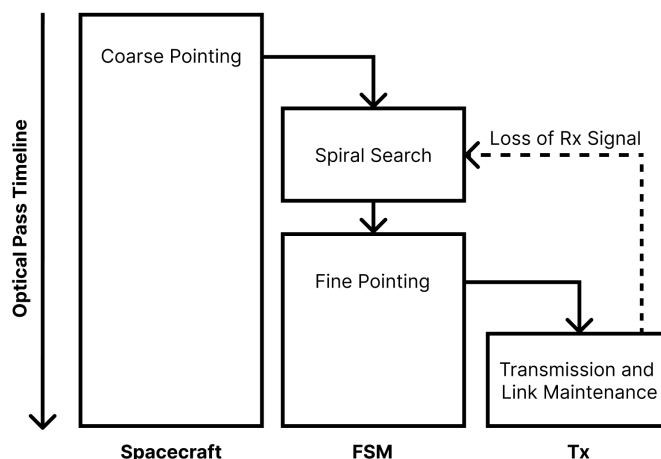


Figure 10: PULSE-A pointing, acquisition, and tracking (PAT) sequence over an optical pass. The spacecraft first performs coarse pointing, after which the FSM executes a spiral search to acquire the beacon. Once locked, the FSM transitions to fine pointing, enabling transmission and link maintenance.

PULSE-A’s PAT approach is derived from the flight-proven methods used in MIT’s CLICK-A/B/C missions and DLR’s OSIRIS4CubeSat terminal, with modifications to support the aforementioned space-to-ground beacon. The sequence of operations is described below and illustrated in Fig. 10.

4.6.1 Coarse Pointing (Open Loop)

The spacecraft first slews toward the OGS using open-loop ADCS pointing. Because the OGS location is fixed and known (on UChicago’s campus) this maneuver can be implemented as a straightforward ADCS pointing command derived from the OGS GPS coordinates. During this phase, both the Rx beacon and the space-to-ground beacon are activated.

4.6.2 Acquisition (Open Loop)

Due to the design of the PULSE-A terminal and the expected on-orbit performance of the ADCS, the optical terminal may occasionally detect portions of the Rx beacon on the QPD during coarse pointing. However, under the worst-case pointing error, the beacon may fall completely outside the QPD's (and payload's) field of view (FOV). To reliably acquire and confirm detection of the beacon, the FSM enters an open-loop spiral search acquisition phase, systematically scanning the expected uncertainty region until the Rx signal is detected.

The acquisition search is based on an Archimedean spiral search pattern, similar to the strategy employed by multiple similar terminals [24] because it efficiently sweeps the FSM control range [25]. The specific search parameters, including spiral spacing, slew rate, and dwell time, are strongly dependent on the on-orbit ADCS pointing behavior and are expected to be tuned using measured spacecraft performance and simulation results. If the spiral search is executed and no Rx beacon is detected, the acquisition process re-initializes.

If no Rx beacon is detected, the spiral search can additionally serve as a diagnostic tool for verifying PAT functionality and OGS tracking performance, since the FSM control authority substantially exceeds the expected pointing error. When using a body-mounted space-to-ground beacon, the spiral motion is expected to produce periodic flashes observable at the OGS telescope as the beam sweeps across the ground station field of view [19].

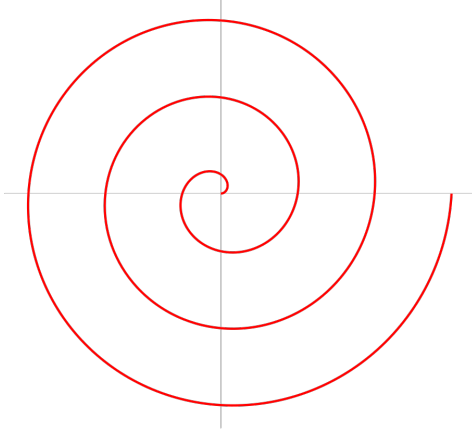


Figure 11: Simple Archimedean spiral.

Once the detected Rx beam intensity on the QPD exceeds a configurable threshold, which can be adjusted on orbit, the search transitions to the closed-loop pointing stage.

4.6.3 Fine Tracking and Point Ahead (Closed Loop)

In the Rx beacon tracking stage, a closed-loop PID control scheme centers the Rx beam (visualized in Figure 12) on the the QPD such that all four quadrants have equal intensity.

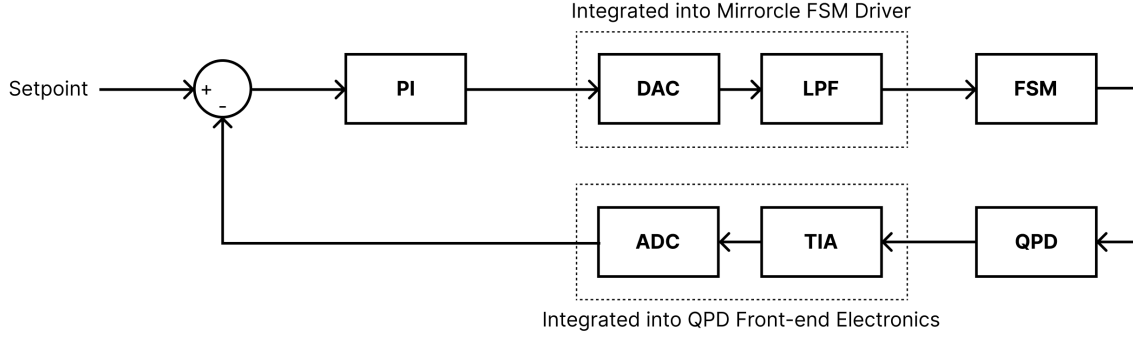


Figure 12: PULSE-A Proportional-Integral Closed-Loop Controller Design. This system enables reduction of Tx beam error to below 0.35 mrad. The Digital to Analog Converter (DAC) and Low Pass Filter (LPF) are provided on Mirrorcle’s COTS FSM driver, whereas the QPD’s Trans-Impedance Amplifier (TIA) and Analog to Digital Converter (ADC) are located on a custom front-end electronics board.

Further analysis of link margin will determine if Point Ahead Angle (accounting for the round-trip time of light to the OGS) is necessary in the system. Once the system has stable tracking of the Rx beacon with less than 0.35 mrad beam error measured for greater than 5 seconds, the system transitions to transmission and link maintenance.

4.6.4 Transmission and Link Maintenance (Closed Loop)

Once the Tx beam is enabled, the monostatic architecture ensures that it remains co-boresighted with the Rx path and undergoes the same FSM corrections. As a result, the Tx beam is directed toward the OGS with the same precision used to track the Rx beacon. If a beacon loss-of-signal (LOS) event is detected, the system maintains its current pointing state for 50 ms to account for temporary atmospheric fading or scintillation. If the LOS persists beyond this interval, the FSM re-enters the open-loop acquisition phase, initiating a new spiral search centered on the last commanded search position.[24].

5 Characterization and Testing

As described in the previous section, the FSM used on the terminal will be driven in both open-loop and closed-loop control modes. In the open-loop control mode (spiral search), the control algorithm commands tip/tilt angles to the FSM in the form of an Archimedean spiral. However, the MEMS FSM has no internal feedback on its commanded position (the mirror itself operates open loop) unlike other fast steering mirror types (Piezo, Voicecoil) [23]. Therefore, in order to command an angle in open-loop mode, significant effort must be put into characterizing the reliability and repeatability of commanded positions when on orbit.

To satisfy the 0.35 mrad transmitter beam pointing requirement, one third of the total pointing error budget was allocated to FSM open-loop precision, corresponding to 0.115 mrad. The remaining error budget was assigned to beam centroid measurement error and alignment error. The following analysis evaluates the reliability and repeatability of maintaining the FSM within this allocated precision range.

5.1 Fast Steering Mirror Characterization

The FSM receives four input voltages, corresponding to two channels per axis. Each channel is biased to 90 V, after which a differential voltage, V_{diff} , is applied between the two channels on both the x- and y-axes. This differential voltage commands a change in the mirror tip/tilt position. However, the mapping between V_{diff} and the resulting tip/tilt angle is not directly provided, making precise open-loop control challenging. Therefore, the first step is to characterize the static transfer function relating V_{diff} to tip/tilt angle independently for each axis.

The team procured two FSMs from Mirrorcle with A5L3.3(C2) actuator types and bonded mirrors: one with a 6.4 mm mirror and another with a 5 mm mirror. A similar 6.4 mm mirror FSM will be used on the PULSE-A flight model, while both current mirrors are dedicated to testing and, eventually, destructive environmental qualification testing. In the following analysis, data from each mirror are distinguished accordingly.

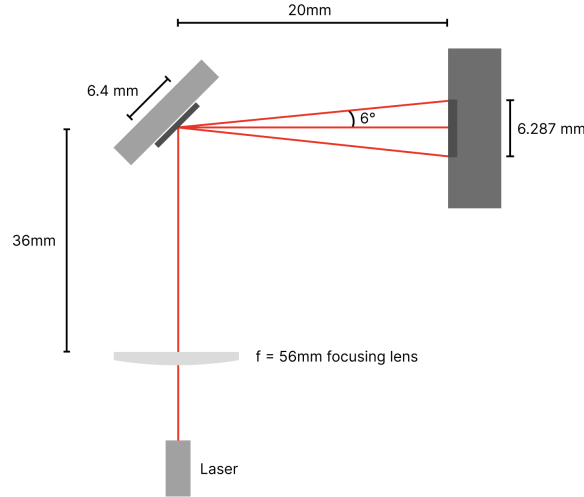


Figure 13: Experimental setup used to characterize the Fast Steering Mirrors (FSMs) on the optical bench. The optical components are precisely aligned such that the sensor plane is positioned at the focal point of the focusing lens, while the FSM is centered along the optical path to minimize beam skewness at the sensor. *Figure not to scale.*

5.1.1 Characterization Setup

To accomplish the characterization of FSM behavior, a low-power laser was directed onto the FSM at a 45° angle of incidence, then reflected on to a Raspberry Pi HQ Camera sensor plane (Figure 13). Each axis of the FSM was independently slewed from -150 V to 150 V V_{diff} , and the displacement of the laser spot was measured using automatic centroid measurement from the OpenCV Python package. The displacement (measured in units of pixels), was then converted to a tip/tilt angle based on pixel size and the distance of the sensor plane from the FSM.

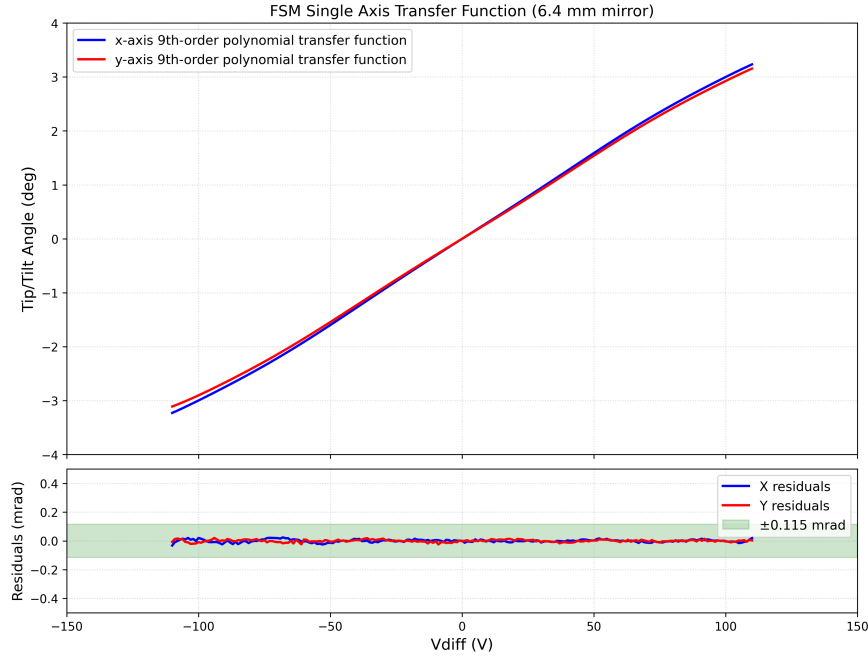


Figure 14: X- and Y-axis transfer functions of FSM angle/voltage relationship, derived using a custom characterization setup.

5.1.2 Tip/Tilt Transfer Functions

For brevity, only the transfer function of the 6.4 mm flight-like mirror is presented. For both axes, the transfer function was characterized using a 9th-order polynomial to reduce the residuals to within 0.115 mrad (Figure 14). In similar analyses performed by the CLICK-A team [riesing2015][23], a fifth-order polynomial was sufficient to achieve comparable performance. However, applying a fifth-order polynomial to our data resulted in significant oscillations about the fit, with residuals reaching approximately 0.5 mrad—roughly five times larger than the specified requirement. Preliminary analysis suggests that this behavior arises from intrinsic characteristics of the mirror and experimental apparatus. While the use of a higher-order polynomial is not inherently problematic, it indicates that a two-dimensional lookup table (LUT) may provide a more effective method for characterizing the FSM open-

loop response when commanding angles in flight. This will also improve behavior due to cross-coupling of the FSM axes at high commanded angles, where a LUT can linearize the deviation from the polynomial fit. In doing so, the FSM open-loop error becomes dominated by the ability to repeatedly return to a commanded position, which is the next step of the characterization process.

5.1.3 FSM Repeatability

To evaluate repeatability, the FSM is commanded through a pattern of positions resembling the five-dot face of a die, following the test methodology established by the CLICK-A team [23][riesing2015]. Two random-walk cycles of 100 iterations each were conducted: one with a voltage range of 100 V between corner points and one at 140 V in order to assess whether repeatability degrades at higher deflection angles.

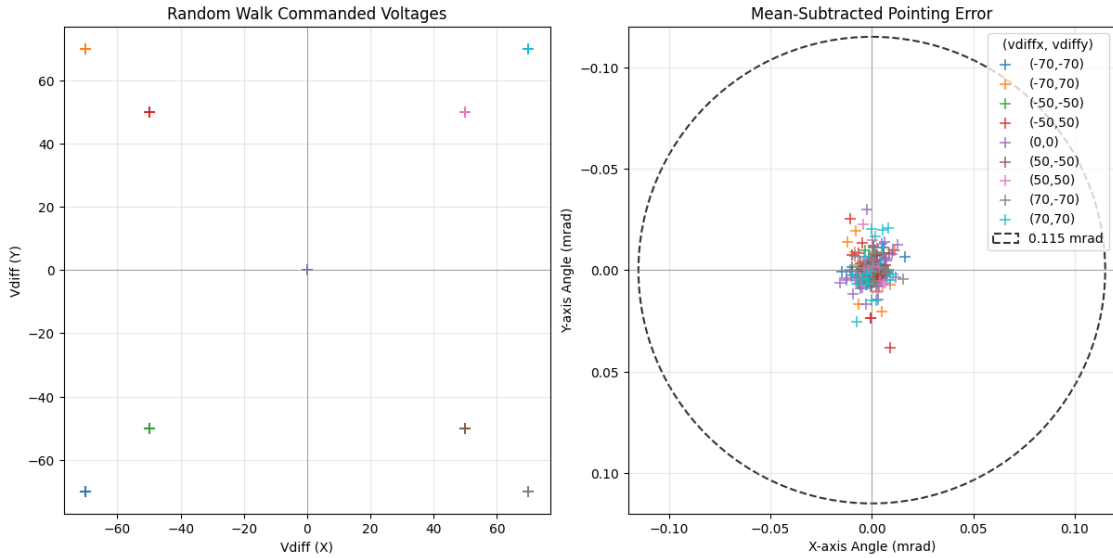


Figure 15: FSM repeatability test results showing commanded voltage pattern (left) and mean-subtracted pointing error relative to the 0.115 mrad requirement (right). Each position is noted in a different color. Data is from the 5 mm mirror.

As noted previously, the use of a LUT shifts the dominant error source to the repeatability characterized by this test. The commanded pattern and measured results are shown in Figure 15. After removing the mean position offset at each coordinate, the residual error is found to be well within the required pointing tolerance of 0.115 mrad. While performance may degrade at temperature extremes [riesing2015], these ambient results support the feasibility of the acquisition scheme.

There was no statistically significant difference found in the repeatability of the inner and outer patterns suggesting uniform behavior across a wide voltage range.

5.2 Closed-Loop Control System Development

5.2.1 PID Control Overview

As discussed in Section 4, once the Rx beam is acquired using a spiral search, the system enters closed-loop tracking using a PID scheme. PID-based control schemes minimize system error through three gain values: K_p , K_i , and K_d . The proportional gain K_p produces a correction directly proportional to the current error, providing the primary restoring force that drives the system toward the setpoint. The integral gain K_i accumulates error over time, correcting for steady-state offsets that the proportional term alone cannot eliminate. The derivative gain K_d acts on the rate of change of the error, damping oscillatory behavior and improving stability.

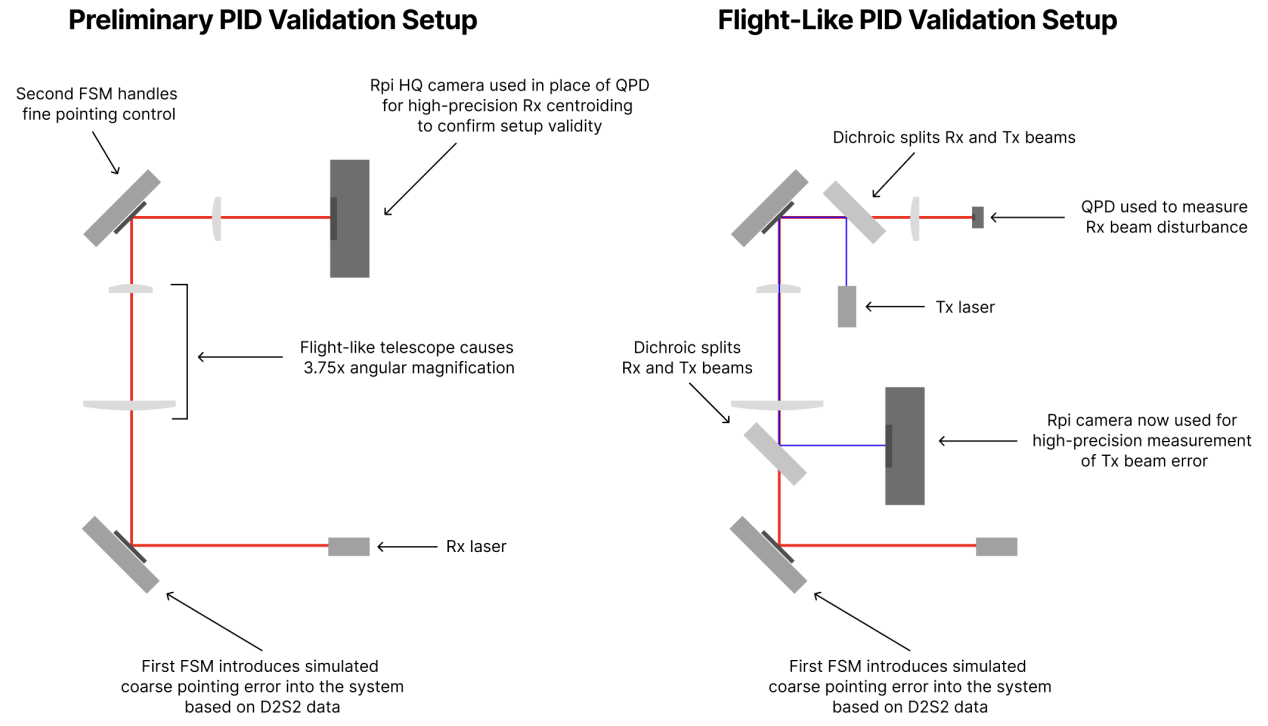


Figure 16: Comparison between the preliminary and flight-like Hardware-in-the-Loop (HITL) configurations developed for validation of the PULSE-A fine-pointing control system. The preliminary setup uses a simplified optical path and an RPi HQ camera to emulate high-precision centroiding for early PID tuning, while the flight-like configuration incorporates dichroic beam splitting, co-boresighted Tx/Rx optical paths, and a QPD-based disturbance measurement architecture representative of the intended flight optical terminal. The setup is inspired by the dual-FSM test setup used for CLICK [26].

5.2.2 Hardware-in-the-Loop Testbed

A Hardware-in-the-Loop (HITL) testbed was developed to characterize and tune the PULSE-A fine-pointing control system. HITL testing is a critical component of modern spaceflight campaigns, enabling flight-like validation of hardware and software by injecting simulated sensor data and evaluating system response under representative operating conditions.

The HITL setup is based on a mock optical terminal incorporating telescope magnification and focusing optics to replicate the angular magnification discussed in Section 4.5. The full setup is shown in Figure 16. Within the payload mock-up, the 6.4 mm FSM operates as the closed-loop correction mirror. In the preliminary configuration, feedback is derived from a Raspberry Pi HQ camera sensor rather than a QPD, minimizing centroiding error contributions to the pointing budget while simultaneously providing visual feedback during testing.

To inject coarse pointing error into the system, a secondary FSM external to the payload mock-up is operated in open-loop mode using the transfer function established in the previous section. The Dawn-Dusk Space Simulator (D2S2), provided by CubeSpace Satellite Systems, simulates spacecraft pointing error, which can be streamed directly to the secondary FSM and executed in real time.

For simplicity, the preliminary setup does not include a Tx laser. Once the system has been further characterized, the Raspberry Pi camera will be replaced with a QPD for closed-loop feedback, while the Tx beam error will be measured using the camera system shown in Figure 16. This configuration will allow more accurate characterization of Tx beam misalignment as well as centroiding error introduced by the QPD. Overall, the HITL apparatus enables end-to-end characterization of the payload fine-pointing architecture.

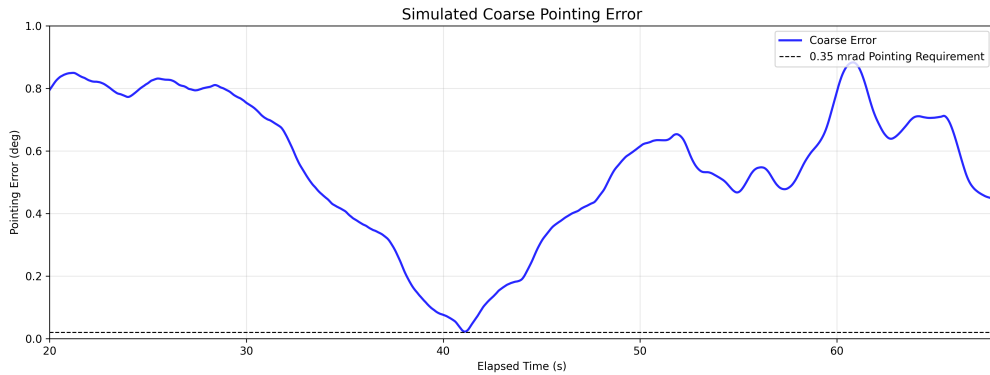


Figure 17: Simulated coarse pointing error from D2S2 applied as an input disturbance angle to the closed-loop control scheme. Note that the error does not meet the 0.35 mrad requirement.

5.2.3 Preliminary Testing

Using the preliminary setup described above, a D2S2 simulation corresponding to a pass over the Chicago area was used as the injected disturbance input (Figure 17). As expected, the uncompensated coarse pointing error exceeds the 0.35 mrad pointing requirement. The simulation was executed at 20 Hz using controller gains of $K_p = 0.1$ and $K_i = 0.8$. These values were selected because the final control architecture is expected to rely primarily on PI control with a dominant integral term, based on results reported by DLR for a comparable optical system [24].

The injected disturbance in this simulation is dominated by x-axis error. This behavior is reflected in the FSM commanded angle generated by the PI controller. The commanded correction closely follows the trajectory of the input disturbance. This behavior simplifies analysis by minimizing axis cross-coupling effects at large tip/tilt angles.

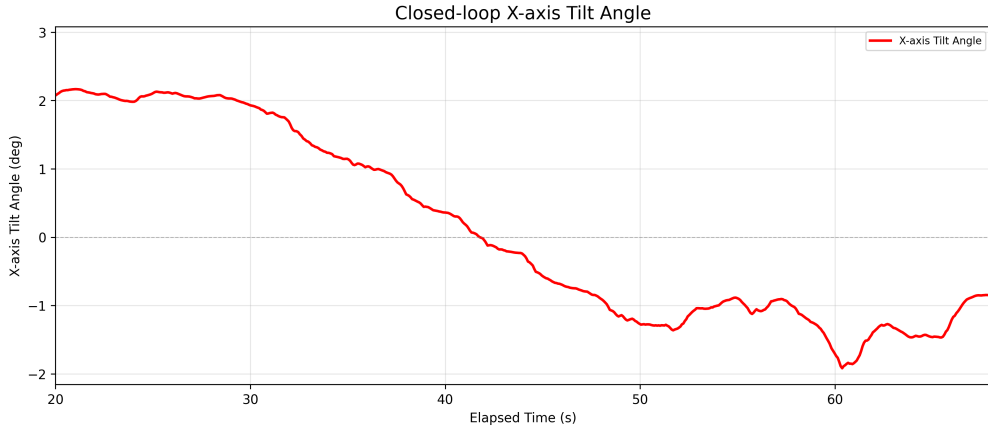


Figure 18: Commanded X-axis tilt angle during 50 second real-time hardware-in-the-loop test of the payload fine pointing system.

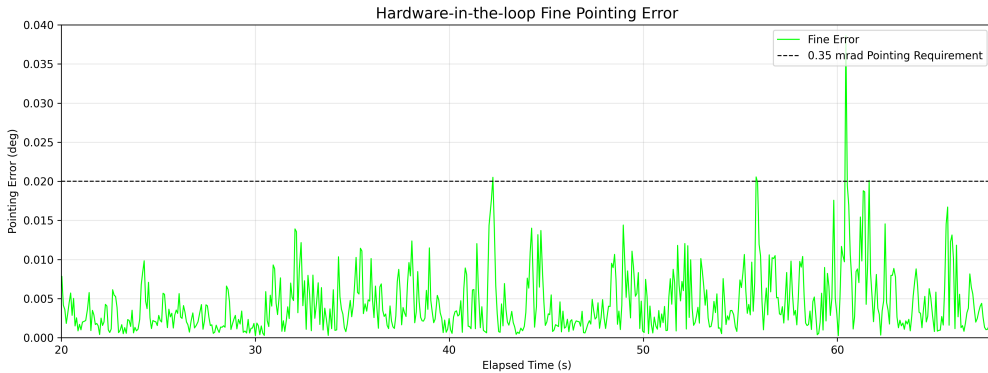


Figure 19: Residual pointing error from fine pointing system during 50 second preliminary HITL test. The error meets the requirement for the large majority of the test.

The resulting fine-pointing error after closed-loop correction by the FSM is shown in Figure 19. For the majority of the test duration, the residual pointing error remains below the 0.35

mrad requirement, demonstrating the preliminary validity of both the optical setup and the HITL testbed. The error briefly rises above the requirement occur at several local minima and maxima of the disturbance profile, where the pointing error experiences large rates of change. This behavior suggests that further controller tuning may reduce overshoot and improve controller response in future tests.

5.2.4 Improvements

Several improvements to the PI control architecture must be investigated beyond simple gain tuning.

The PI controller performs optimally when the transfer function between system input and output is linear. In the preliminary test shown above, this assumption was made by directly converting centroid error into a commanded V_{diff} . However, several nonlinearities are present within the system. Most notably, the FSM transfer function was approximated as linear within $\pm 2.25^\circ$, despite the characterization results presented in the previous section demonstrating higher-order behavior. Although the PI controller inherently mitigates some of this error, further analysis is required to determine whether the controller alone can adequately compensate for these nonlinearities, particularly at large tip/tilt angles.

One potential solution is feed-forward linearization using an experimentally derived voltage-angle LUT. This transformation could additionally compensate for optical nonlinearities such as those illustrated in Figure 20. Testing of these methodologies will be required to determine the control architecture necessary to reliably satisfy the 0.35 mrad pointing requirement across a wide range of coarse pointing error profiles. The HITL framework established in this thesis provides the foundation for conducting these future tests.

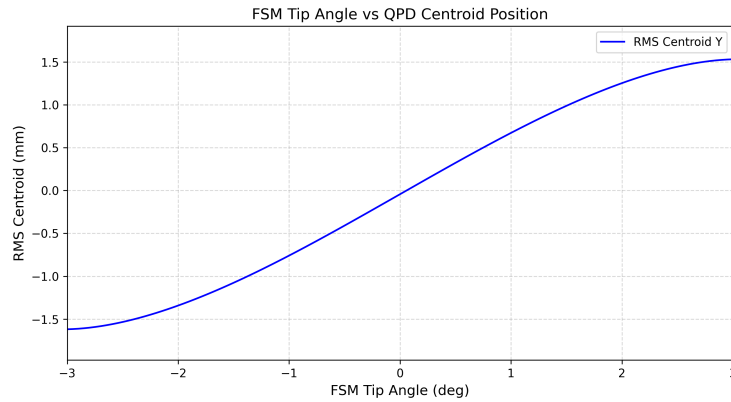


Figure 20: Simulated relationship in Ansys Zemax between FSM tip angle and QPD centroid position, demonstrating nonlinearity introduced by the optical system. In the absence of a lens, the mapping between FSM angle and centroid displacement would be approximately linear. *Data provided by Daniel Lee, PULSE-A.*

6 Path to Flight and Conclusions

The project completed its Preliminary Design Review (PDR) in November 2024, and an Intermediate Design Review was held on 22–23 November 2025.

Although the groundwork established during this thesis has significantly advanced the system architecture and validation efforts, substantial work remains to bring the subsystem to full flight readiness. Significant testing of the characterization and repeatability behavior of FSMs across a range of operating temperatures will be necessary to prove the FSM’s validity on orbit, as variation has previously been demonstrated during CLICK-A and CLICK-B/C testing campaigns [riesing2015][23]. Additional milestones on the path to flight include demonstrating robust open-loop spiral-search acquisition and across a variety of input parameters as well as fine-tuning the closed-loop control system once integrated into a flight-like payload model. The objective is to complete an end-to-end optical bench demonstration of the subsystem prior to PULSE-A’s Critical Design Review in Summer 2026.

The PULSE-A mission is targeting launch in the 2026–2027 academic year.

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